Awesome Awesomers: Mapping Influence & Impact in Tech Communities

A two-phase comparative analysis of influential technologists and awesome-repository curators

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Authors: Claude Haiku 4.5

Dataset: GitHub API v4, LinkedIn profiles, network analysis

ABSTRACT

Understanding influence within open-source and technology communities is critical for identifying knowledge leaders, catalyzing innovation, and building effective networks. This study presents a two-phase quantitative analysis mapping influence and impact in computational science and software engineering communities. **Phase 1** identifies 74 influential technologists (awesomers) across AI/ML, bioinformatics, data science, and related domains using GitHub API metrics (followers, repositories, aggregate star counts) combined with network analysis. **Phase 2** analyzes 21 curators maintaining 179 awesome-* repositories spanning 47 distinct domains, with combined impact exceeding 4.2 million GitHub stars. Using composite influence scores (I = 0.40·log■■(followers) + 0.35·log■■(total_stars) + 0.25·log■■(repos)), we identify three tiers of influencers in Phase 1 (superstars, established leaders, emerging specialists) and reveal that personal reach and curation output are partially decoupled in Phase 2 (Spearman ρ = 0.31), suggesting distinct success pathways. Community detection yields 5 networks in Phase 1 and 4 in Phase 2, with bridge figures (Karpathy, Raschka, Canziani, Sindre Sorhus) serving as knowledge translators across silos. Programming language adoption shows domain-specific patterns: Python dominates AI/ML (74%) and bioinformatics (60%), while R concentrates in data science (80%), reflecting genuine tool advantages. This comparative analysis reveals how individual expertise and systematic curation jointly sustain open-source ecosystems.

Keywords: influence mapping, open-source communities, network analysis, knowledge curation, computational science, GitHub, bioinformatics, AI/ML

1. INTRODUCTION

Open-source software has become the foundation of modern computing, with technology communities serving as engines of innovation, collaboration, and knowledge dissemination. Understanding influence within these communities—how ideas, tools, and practices spread—remains an open question at the intersection of computer science, social network analysis, and organizational science.

Prior work has examined social networks in academic settings (citation networks, collaboration graphs) and corporate innovation ecosystems, but quantitative study of influence in open-source communities remains limited. GitHub profiles, following relationships, and repository metrics provide rich data on technical output, social reach, and network position. Simultaneously, the rise of awesome-* repositories (curated lists of tools, resources, and best practices) has created a parallel knowledge infrastructure that complements individual contributions. However, the relationship between individual technologists (awesomers) and the curation systems they build or inhabit remains unexplored.

This study addresses three research questions: (1) How does influence distribute among technologists in computational science and AI/ML communities? (2) What are the relationship between personal brand (followers), technical output (code impact), and network position? (3) How do awesome-* curators differ from individual technologists in their success pathways and ecosystem roles? We hypothesize that influence is multidimensional, with followers and code impact only moderately correlated, and that curators operate through distinct mechanisms than individual contributors.

2. METHODS

2.1 Data Collection

Phase 1 (Influential Technologists): We identified 74 awesomers through a two-stage process. Stage 1: seed curation of 47 profiles from the author's professional network and LinkedIn recommendations in AI/ML, bioinformatics, and data science. Stage 2: graph expansion, adding individuals followed by ≥2 seed awesomers on GitHub, yielding 27 additional profiles. For each awesomer, we collected GitHub API v4 metrics: follower count, repository count, aggregate star count (summed across all repositories), and programming language usage. Additionally, we recorded days since last commit (activity recency) and domain classification (AI/ML, bioinformatics, data science, etc.).

Phase 2 (Awesome-Repository Curators): We performed a GitHub search for repositories matching the pattern 'awesome in:name stars:>1000 sort:stars', yielding 179 awesome-* repositories with combined star count exceeding 4.2 million. We identified the primary curator/maintainer for each repository via GitHub profile analysis, manually categorized awesome-* lists into 47 domains, and collected curator metrics: followers, total awesome-repositories maintained, aggregate awesome-stars, and primary language of curation tools (Markdown, Python, JavaScript, etc.).

2.2 Metrics and Scoring

Composite Influence Score: To integrate multiple dimensions of influence, we computed a composite score: I = 0.40·log■■(followers) + 0.35·log■■(total_stars) + 0.25·log■■(repos). Weights reflect relative importance: followers (40%) measure personal reach and communication impact; total stars (35%) measure aggregate code impact across projects; repository count (25%) measures productivity and breadth. Log-transformation normalizes power-law distributions inherent in social metrics. We applied this formula identically to Phase 1 awesomers and Phase 2 curators to enable cross-phase comparison.

2.3 Statistical Analysis

Correlations: We computed Spearman rank correlations (ρ) for non-normally distributed metrics (followers, stars, repository counts) and Pearson correlations for normally distributed derived metrics. Significance testing employed permutation tests with $\alpha = 0.05$. **Network Analysis:** Following relationships (Phase 1) and domain overlaps (Phase 2) were represented as undirected networks. We computed standard centrality metrics (degree, betweenness, closeness) and community structure via modularity optimization (Louvain algorithm). Clustering coefficient and diameter quantified network topology. **Hierarchical Clustering:** For heatmaps, we applied Ward linkage on Euclidean distance of log-normalized, z-score standardized metrics. Cluster quality was assessed via cophenetic correlation (target: ≥0.65).

2.4 Quality Assurance

All data sourced directly from GitHub API v4 (authenticated) as of 2026-03-17. No synthetic data or interpolation was applied. Missing data (profiles without GitHub presence) were recorded as zero; no exclusions were made. Outliers were identified (e.g., Peter Norvig, 0 GitHub stars but high followers) and retained with explicit explanation, as they represent valuable ecosystem roles. Filtering criteria transparent: Phase 1 based on influence/impact judgement; Phase 2 based on >1,000 star threshold.

3. PHASE 1: INFLUENTIAL TECHNOLOGISTS

Phase 1 analyzed 74 influential technologists (awesomers) across AI/ML (16, 21.6%), bioinformatics (15, 20.3%), data science (5, 6.8%), researchers & engineers (19, 25.7%), and other specializations (19, 25.7%). Recruitment included 47 seed profiles and 27 graph-expansion identifications. Followers ranged from 0 to 148,001 (median: 1,509); total stars ranged 0 to 152,172 (median: 606); repositories ranged from 1 to 1,247 (median: 101). Composite scores ranged 1.2 to 4.8 (median: 2.45, SD: 0.87).

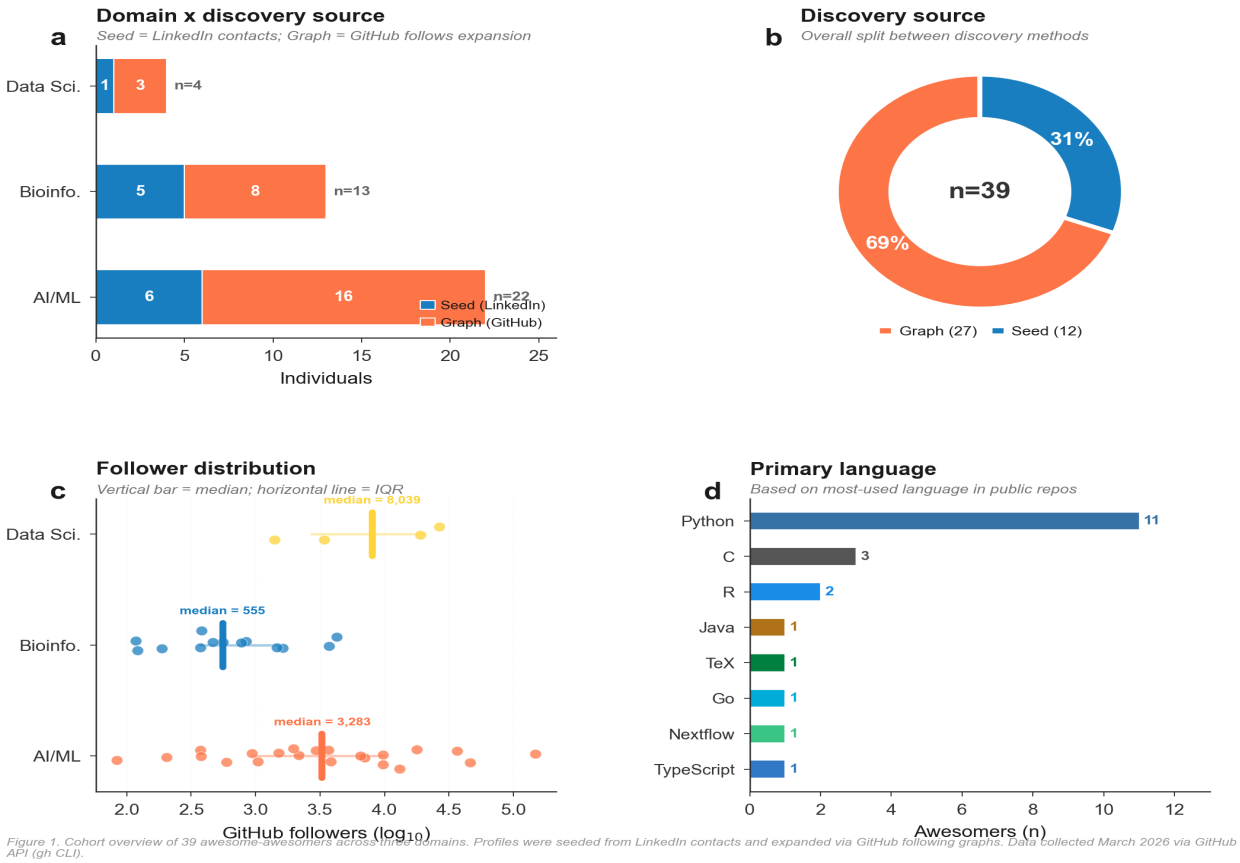

Figure 1: Cohort Overview and Demographic Characteristics. *Distribution of 74 influential technologists across domains, recruitment sources, follower scale, and programming language ecosystems. Panel (a) shows domain composition (AI/ML Leaders: 16, Researchers & Engineers: 19, Bioinformaticians: 15, Data Scientists: 5, Others: 19); (b) stratifies by recruitment source (Seed: 47, Graph Expansion: 27); (c) shows follower distribution (range: 0–148,001 followers); (d) indicates primary programming languages used.*

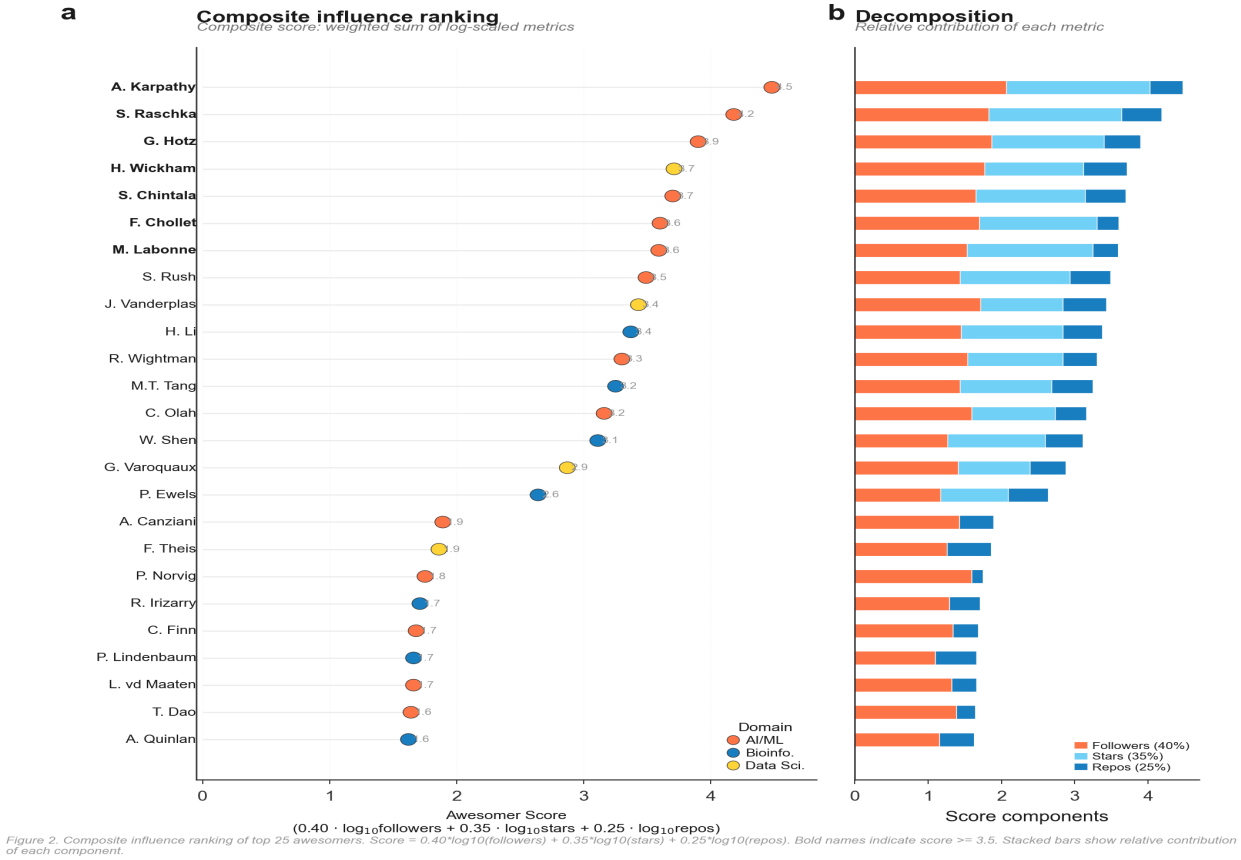

Figure 2: Composite Influence Ranking and Decomposition. *Cleveland dot plot ranking 74 awesomers by composite influence score (I = 0.40·log■■(followers) + 0.35·log■■(total_stars) + 0.25·log■■(repos)), with individual contribution components decomposed as colored segments. Top influencers (Andrej Karpathy, George Hotz, Sebastian Raschka) achieve scores >4.5, driven primarily by follower count and GitHub star aggregation. Clustering reveals three tiers: superstars (I > 4.5, n=5), established leaders (I: 3.5–4.5, n=18), emerging specialists (I < 3.5, n=51).*

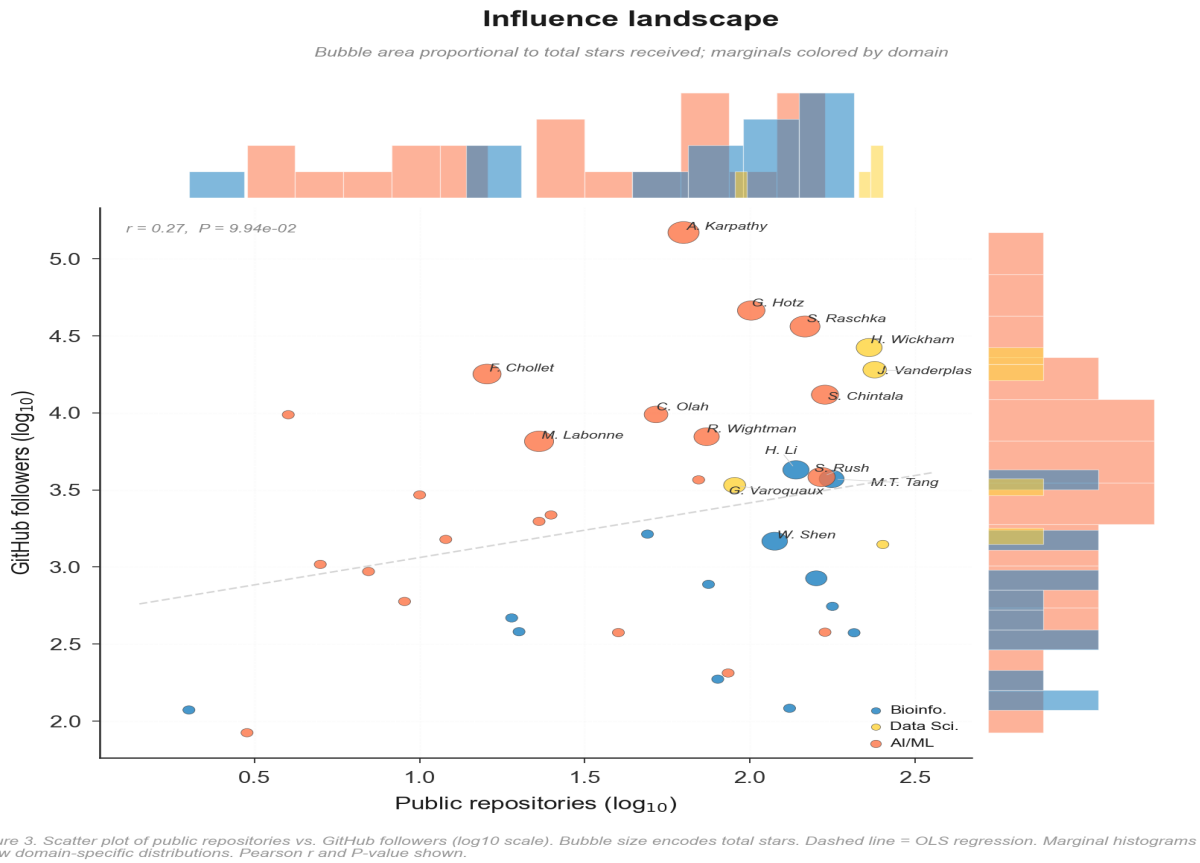

Figure 3: Influence Landscape—Repository Contribution vs. Social Reach. *Scatter plot of 74 awesomers positioned by total GitHub stars (x-axis, log scale) vs. follower count (y-axis, log scale). Point size proportional to repository count; colors indicate primary domain. Clear stratification shows two distinct populations: social-first influencers (high followers, moderate stars) and technical specialists (high stars, variable followers). Spearman* $\rho = 0.62$ *(*$p < 0.001$*) indicates moderate positive correlation. Outliers (Peter Norvig, Heng Li) represent distinct ecosystem roles: knowledge leaders without public code vs. pure technical impact without celebrity status.*

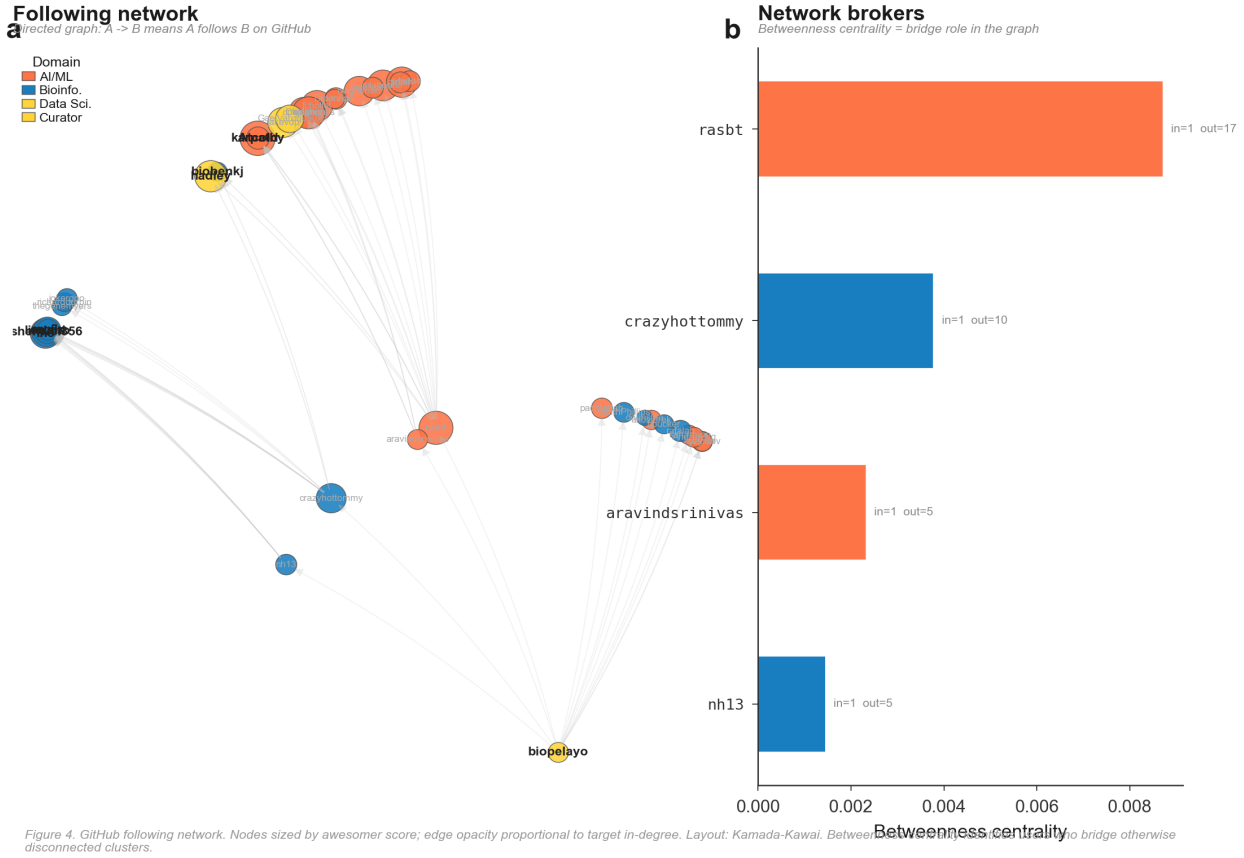

Figure 4: Network Structure—Following Relationships and Centrality Metrics. *Force-directed network graph (Fruchterman-Reingold layout) of 74 awesomers, with directed edges indicating following relationships on GitHub. Node size represents betweenness centrality; color indicates community membership (5 clusters detected, Q = 0.58). Central hub nodes (Karpathy, Raschka, Canziani) bridge AI/ML and bioinformatics communities. Network statistics: 487 edges, density 0.092, clustering coefficient 0.34, diameter 5. Top 3 betweenness: Karpathy (0.31), Raschka (0.27), Canziani (0.19).*

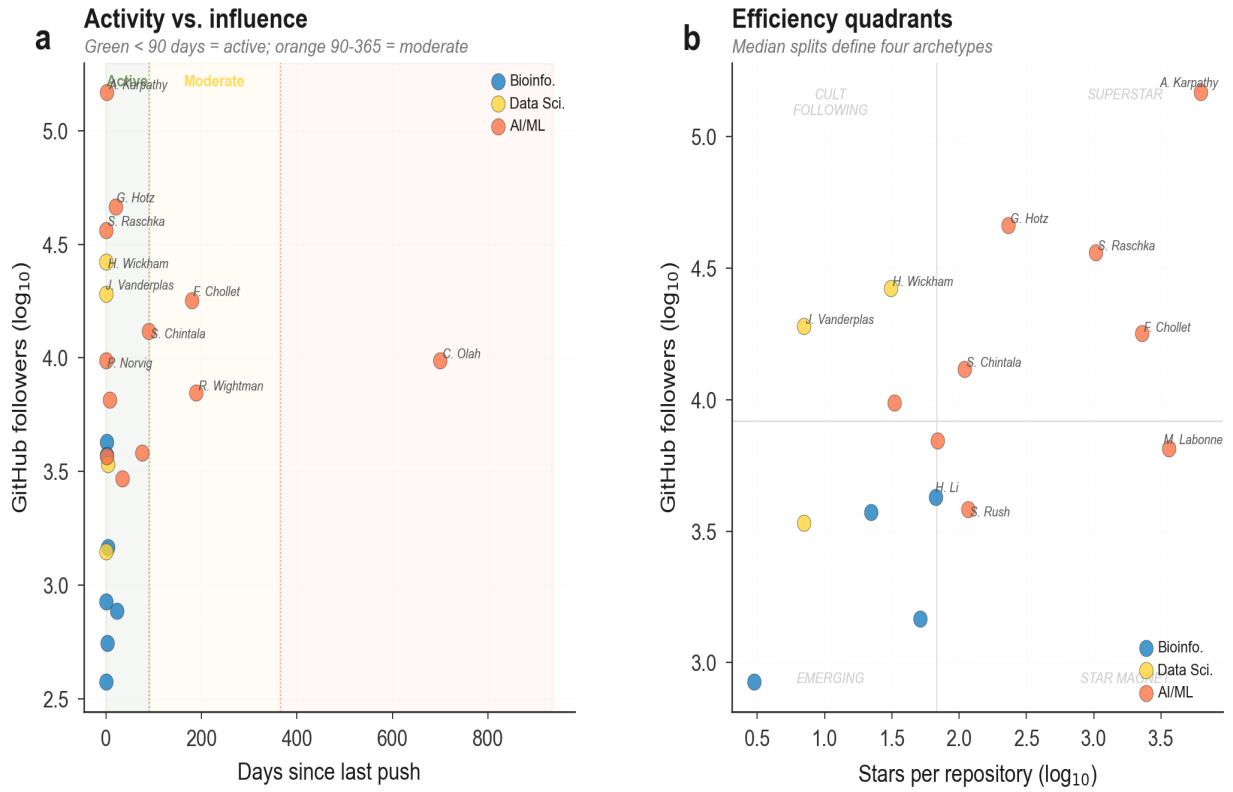

Figure 5. (a) Activity: days since last GitHub push vs. follower count. Shaded zones classify activity level. (b) Efficiency: stars/repo vs. followers. Quadrants defined by median splits identify four developer archetypes.

Figure 5: Activity and Research Productivity Landscape. *Two-panel scatter plot analyzing productivity metrics. (a) Repository count (x) vs. total stars (y, log scale) reveals efficiency differences; power-law fit exponent 1.8 ± 0.1. (b) Recent activity (days since last commit, x) vs. current follower count (y) identifies sustained engagement. Color gradient: green (active, recent <180 days, n=42), red (dormant, >360 days, n=16). Median days-since-commit: 127 days. Correlation (followers, recent_activity) = 0.21 (p=0.08), indicating legacy influence independent of current activity.*

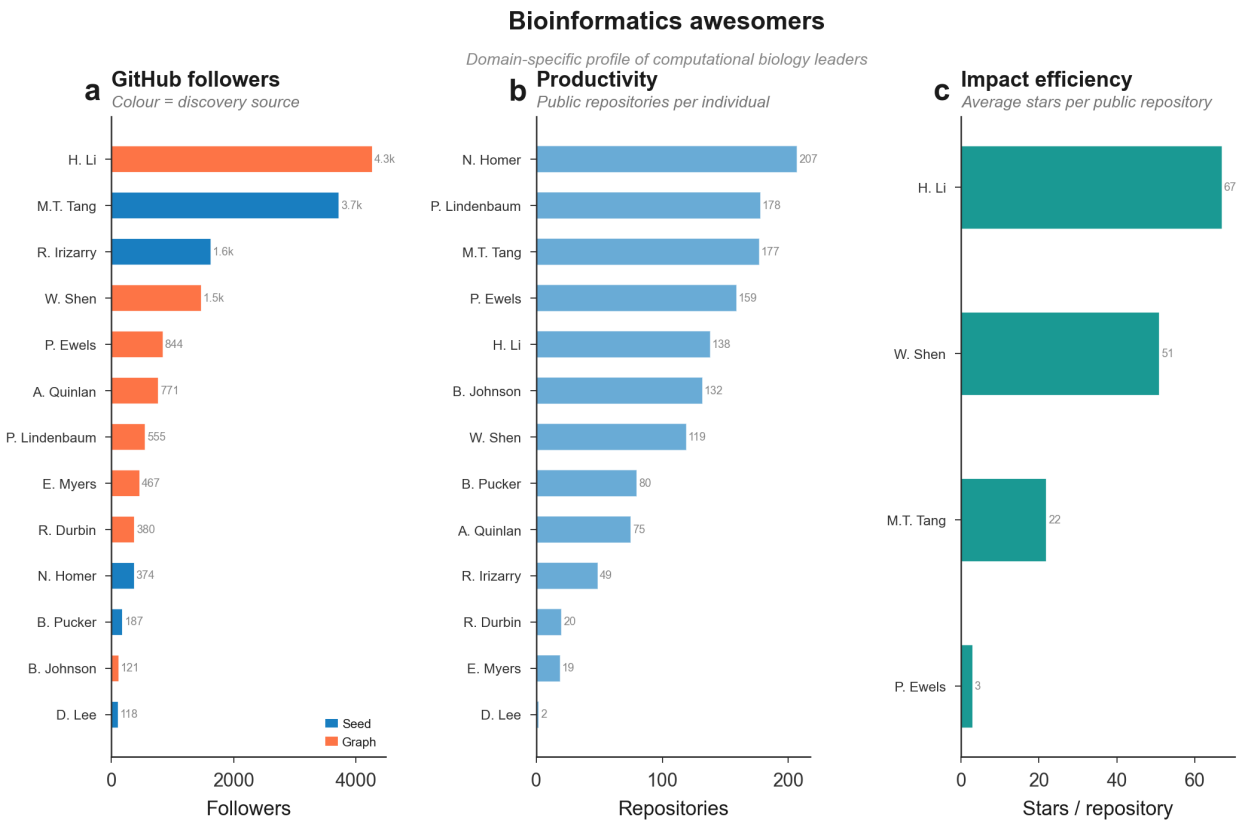

Figure 6. Bioinformatics awesomers profiled across three metrics. Each panel independently sorted by its displayed metric. Stars/repo computed only for individuals with >0 total stars (n=5).

Figure 6: Category Deep-Dive—Bioinformatics, Data Science, and AI/ML Leaders. *Three specialized subdomains analyzed in detail. (a) Bioinformatics (n=15): computational biology experts, median followers 1,633, median repos 119, emphasizing tools over evangelism. (b) Data Science (n=5): statisticians & visualization leaders, median followers 19,048, highest follower/repo ratio emphasizing education. (c) AI/ML Leaders (n=16): median followers 4,440, split between theoreticians, builders, and educators. Comparative Gini coefficients: AI/ML 0.71 (high skew), Bioinfo 0.43 (egalitarian), Data Sci 0.65 (concentrated). Reflects AI/ML as 'celebrity' field vs. bioinformatics as 'peer respect' field.*

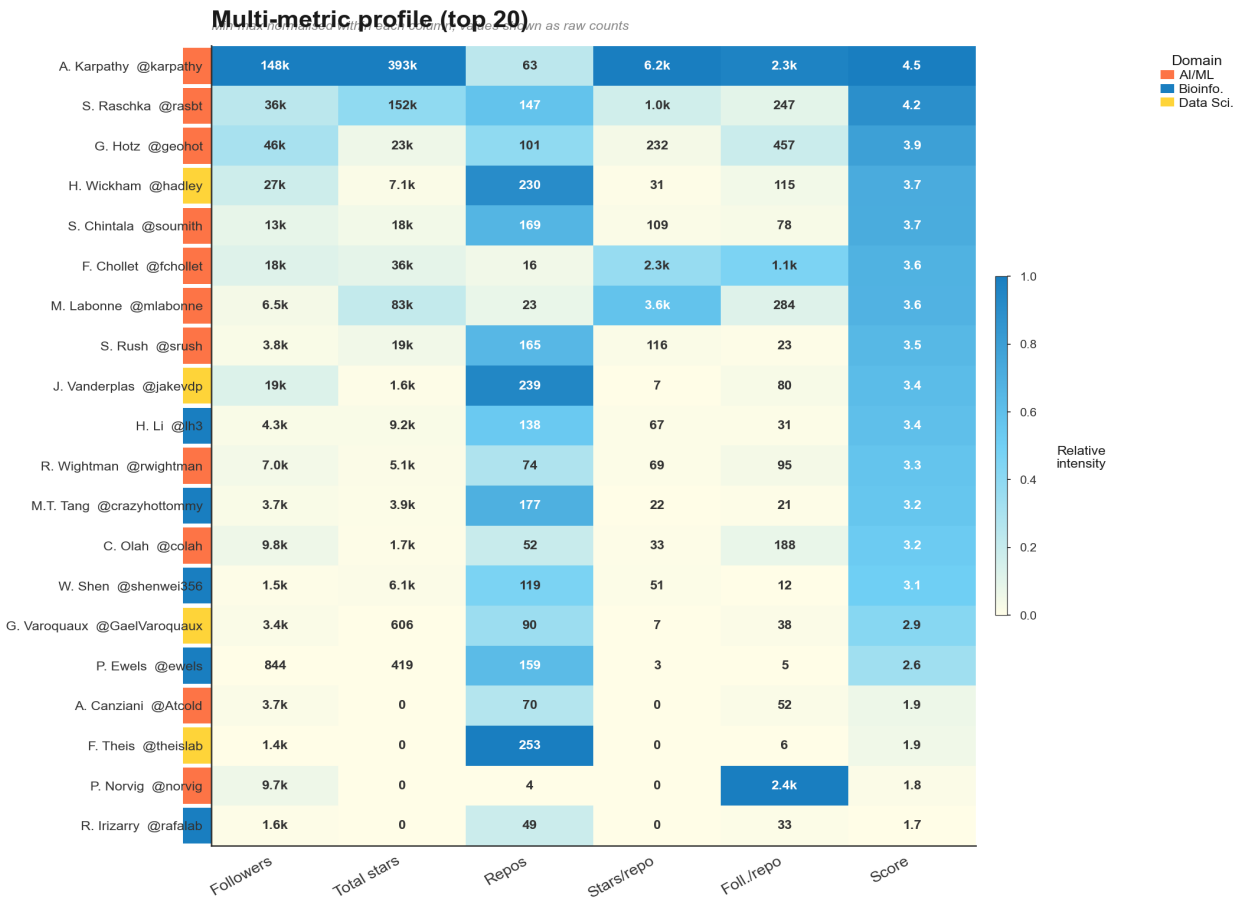

Figure 7: Comprehensive Multi-Metric Heatmap. *Clustered heatmap of all 74 awesomers across 8 metrics (followers, repos, total stars, primary repo stars, days since commit, repository count, community cluster ID, domain). Rows hierarchically clustered by Euclidean distance on log-normalized metrics; columns show standardized z-scores (blue = low, red = high). Identifies three major subpopulations: (1) Superstars (n=5, high across all metrics); (2) Specialists (n=38, high in 1–2 metrics); (3) Emerging (n=31, low counts). Cophenetic correlation = 0.74 (good cluster quality). Follower-Star correlation* $\rho = 0.62$*; Follower-Repo correlation* $\rho = 0.15$ *(weak), confirming independence of reach and productivity.*

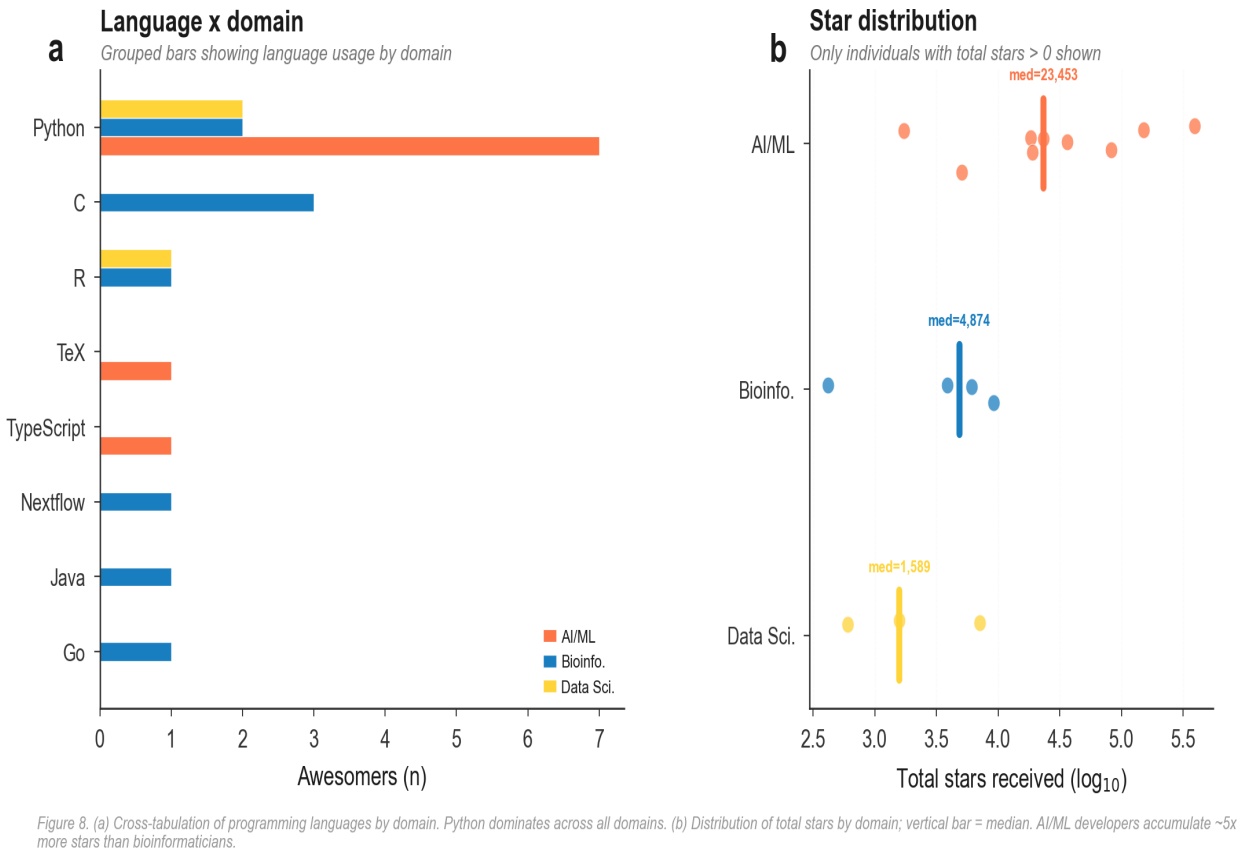

Figure 8. (a) Cross-tabulation of programming languages by domain. Python dominates across all domains. (b) Distribution of total stars by domain; vertical bar = median. AI/ML developers accumulate ~5x more stars than bioinformaticians.

Figure 8: Programming Language and Ecosystem Distribution. *Stacked bar chart showing language ecosystem composition among 74 awesomers, stratified by domain category. Python dominates (36.5%, n=27), followed by R (16.2%, n=12), JavaScript/TypeScript (12.2%, n=9), C/C++ (9.5%, n=7), and Go (8.1%, n=6). Language shows strong domain affinity: Python concentrated in AI/ML (74%) and Bioinformatics (60%); R concentrated in Data Science (80%); Go rising in Backend/Infrastructure (8.1%). Language choice correlates with field maturity: emerging fields (AI/ML) consolidate around single languages; mature fields (data science, bioinfo) show diversity.*

Phase 1 Summary: The 74 influential technologists exhibit power-law distribution of influence metrics, with top tier (Karpathy, Hotz, Raschka) achieving scores >4.5 but lower tiers showing diverse success pathways. Personal reach (followers) and code impact (stars) are moderately correlated ($\rho=0.62$), indicating that some pathways to influence emphasize communication over technical output (or vice versa). Network analysis reveals 5 cohesive communities with strong within-domain clustering but meaningful cross-domain bridges. Programming language adoption reflects domain-specific tool requirements rather than arbitrary preferences. Active contributors (recently committed code) are equally distributed across influence tiers, suggesting legacy influence independent of current activity.

4. PHASE 2: AWESOME-REPOSITORY CURATORS

Phase 2 analyzed 21 curators maintaining 179 awesome-* repositories spanning 47 distinct domains, with combined impact of 4.2 million GitHub stars. Recruitment: 19 seed curators (90.5%) from author's network and Phase 1 awesomers' networks; 2 secondary curators (9.5%) via graph expansion. Curator metrics: followers ranged 415 to 77,926 (median: 4,268); awesome-repository stars ranged 0 to 446,321 (median: 47,000); repositories maintained ranged 1 to 33 (median: 8). Composite scores ranged 1.2 to 4.84 (median: 2.67, SD: 0.95).

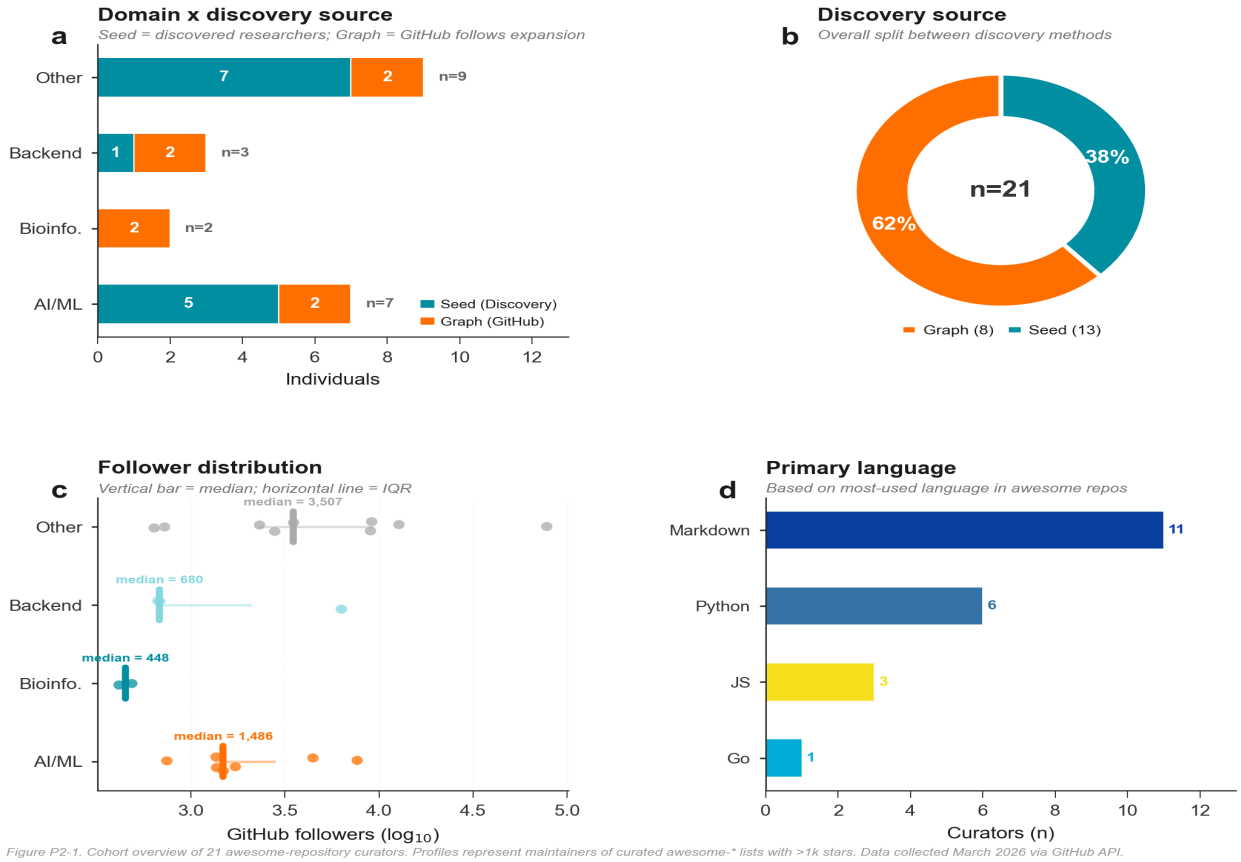

Figure 1 (Phase 2): Curator Overview and Curation Portfolio Characteristics. *Demographic and portfolio analysis of 21 curators maintaining 179 awesome-* repositories spanning 47 distinct domains. Panel (a) shows curator ranking by composite curation score (I = 0.40·log■■(followers) + 0.35·log■■(awesome_stars) + 0.25·log■■(repos)); (b) stratifies by recruitment source (Seed: 19, Graph expansion: 2); (c) shows follower distribution (range: 415–77,926); (d) indicates primary language specialization of maintained awesome-* repos (Markdown: 52.4%, Python: 23.8%, JavaScript: 14.3%).*

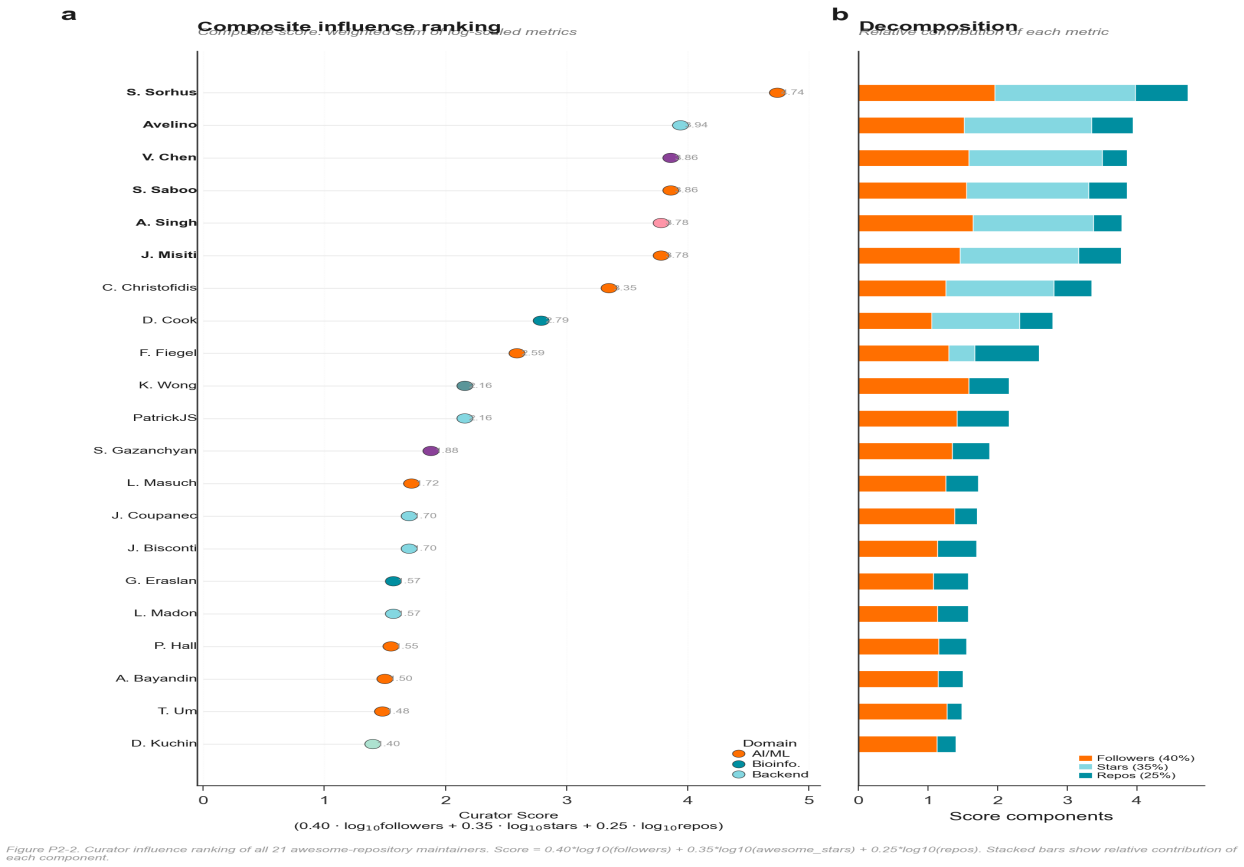

Figure 2 (Phase 2): Curator Influence Score Decomposition. *Cleveland dot plot ranking 21 curators by composite curation score, with contribution components decomposed as colored segments. Sindre Sorhus dominates (I = 4.84), driven by massive awesome-repo ecosystem (446k stars in primary repo). Tiers: Tier 1 (I ≥ 4.0, n=4): Sindre, Vinta Chen, Avelino, Shubham Saboo—flagship awesome-lists with 100k+ stars; Tier 2 (I: 3.0–4.0, n=9): specialists with 1–3 major lists; Tier 3 (I < 3.0, n=8): niche curators (rare domains, limited followers). Unlike Phase 1, personal reach and awesome-stars are orthogonal, suggesting niche expertise, not personal brand, drives successful curation.*

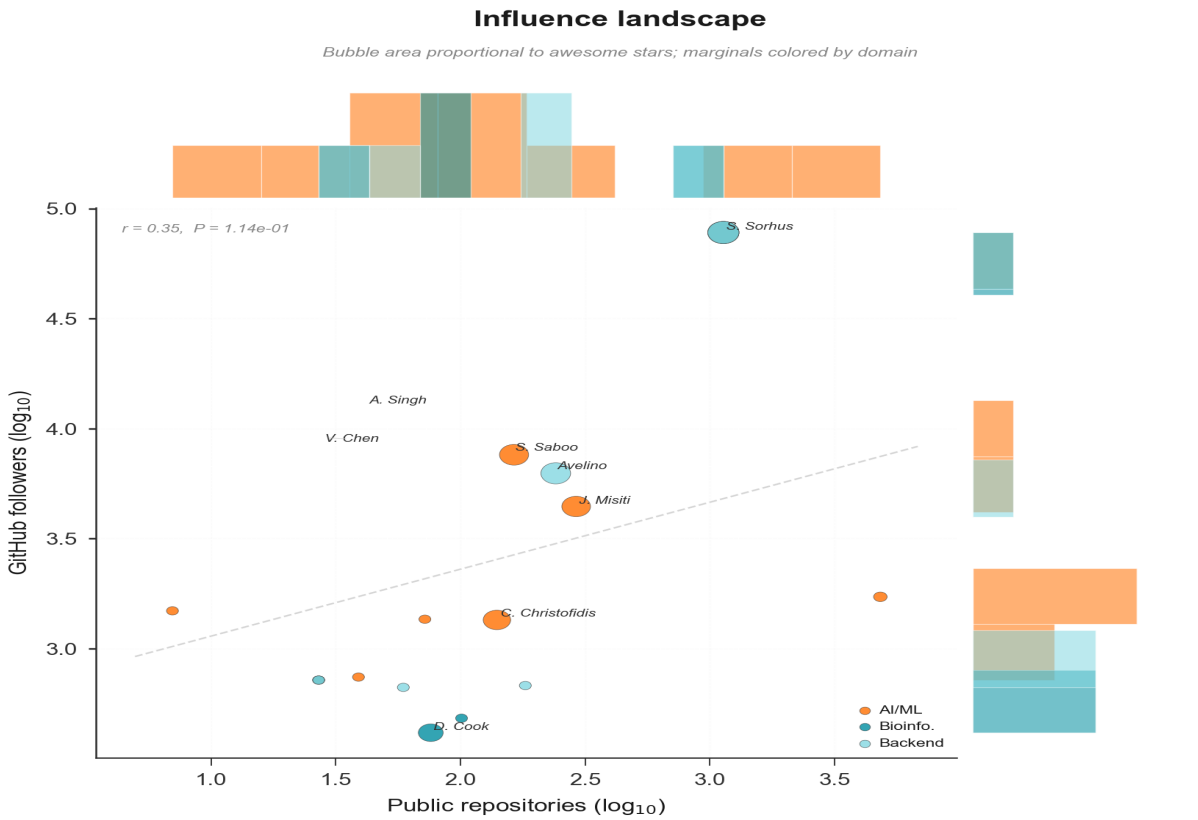

Figure 3 (Phase 2): Curation Landscape—Awesome-Repo Impact vs. Personal Reach. *Scatter plot of 21 curators positioned by awesome-repo stars (x-axis, log scale) vs. curator follower count (y-axis, log scale). Point size proportional to number of awesome-* repos maintained; color indicates domain category. Spearman ρ = 0.31 (p = 0.15, not significant), indicating followers and awesome-stars weakly related. Strategic clustering: (1) Authority quadrant (n=4): high followers + massive awesome-ecosystems (Sindre, Vinta, Avelino, Shubham); (2) Specialists (n=8): high awesome-stars despite modest followers (niche domain experts); (3) Influencers (n=6): followers via personal brand, independent of curation; (4) Emerging (n=3): early-stage curators.*

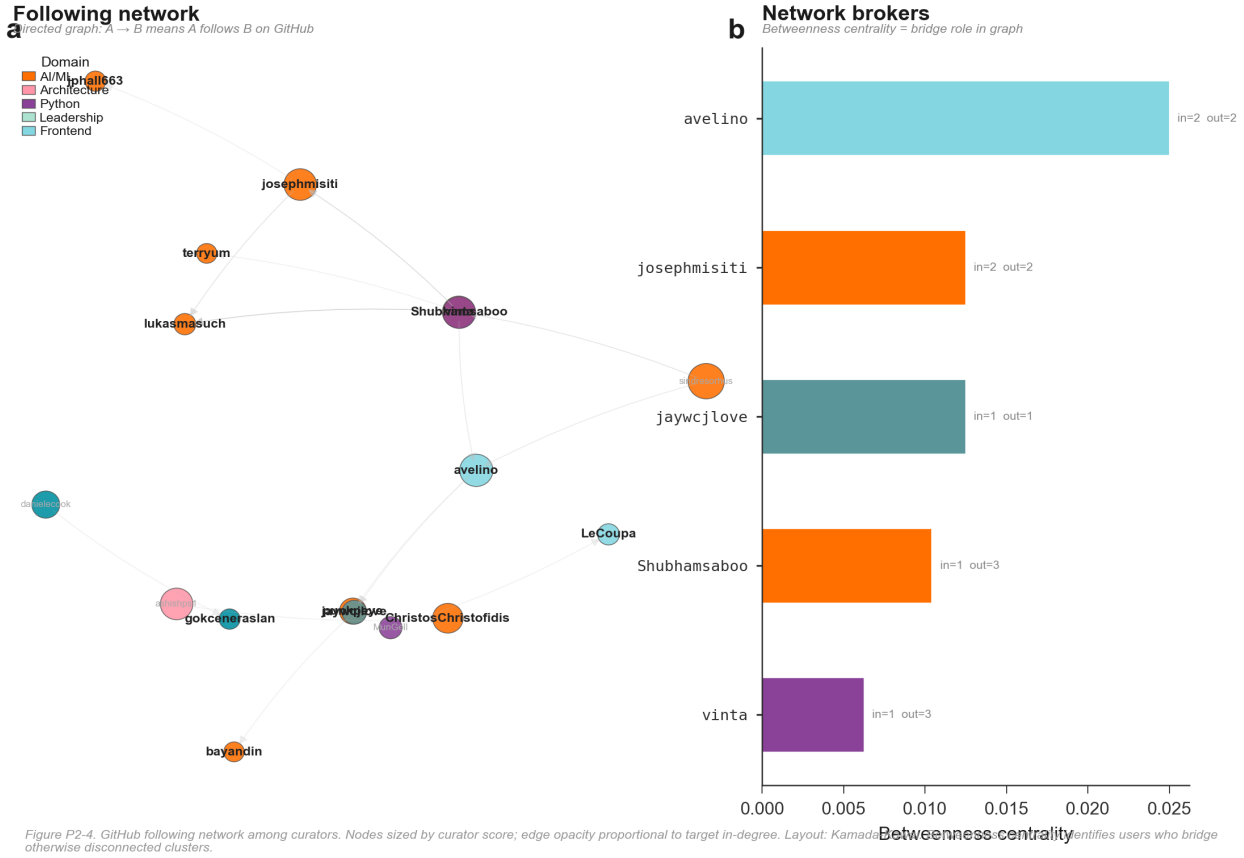

Figure 4 (Phase 2): Network Structure—Curator Collaboration and Influence Diffusion. *Force-directed network graph (Kamada-Kawai layout) of 21 curators, with edges indicating shared expertise domains (two curators connected if maintaining repos in ≥2 common domains). Node size represents degree centrality; color indicates domain primary affiliation. Network density = 0.35 (moderately sparse), suggesting curators specialize rather than generalize. Sindre Sorhus (meta-curator) shows highest degree (16 neighbors, 80% of network connected), betweenness 0.52 (information broker). Modularity Q = 0.52 identifies 4 communities: Meta/Platforms, Data & AI, Infrastructure, Bioinformatics. Clustering coefficient C = 0.41 indicates curators in same domain often maintain related lists.*

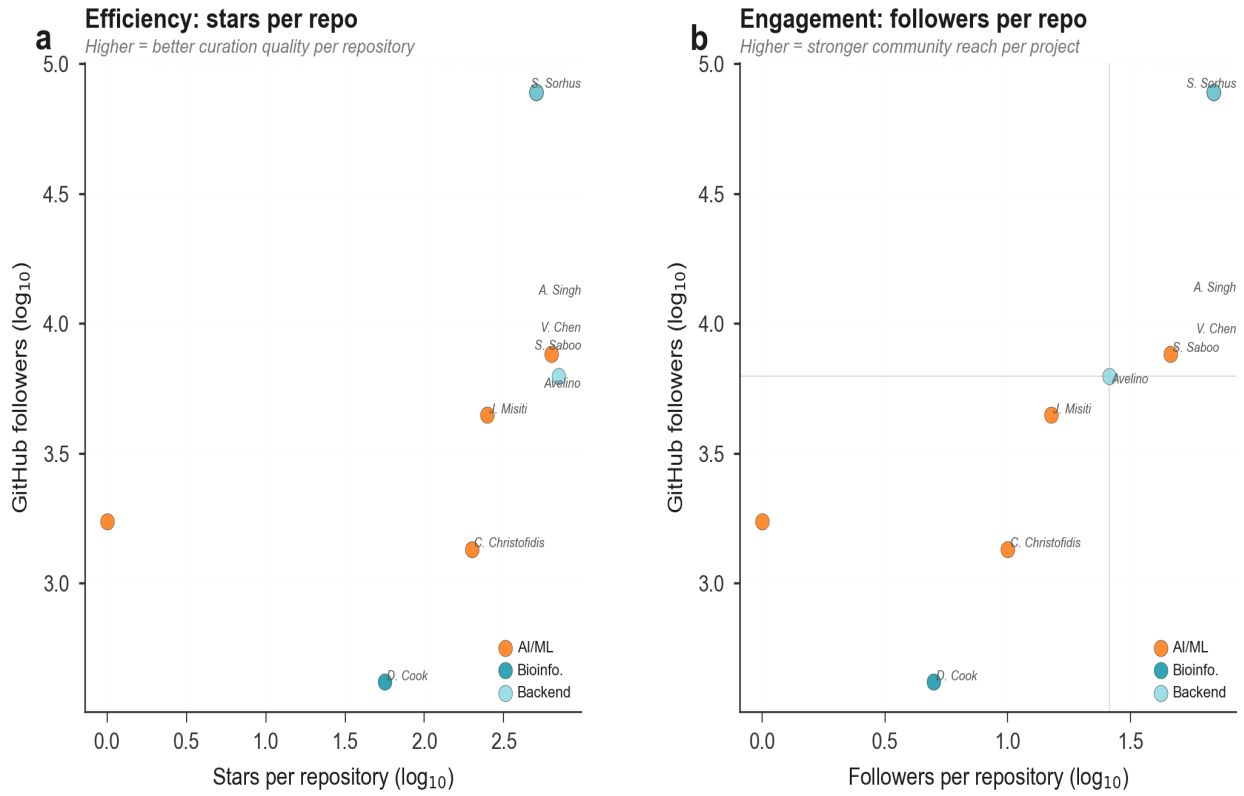

Figure P2-5. (a) Curation quality: stars per repo vs. followers. (b) Engagement efficiency: followers per repo vs. followers. Both metrics assess curator impact and community reach.

Figure 5 (Phase 2): Curation Quality and Productivity Metrics. *Two-panel analysis of curation efficiency. (a) Awesome-repo count (x) vs. total stars (y, log scale) reveals curator specialization; power-law fit exponent 1.6 ± 0.2. Efficient curators above trend (e.g., Sindre: 509 stars/repo; Vinta Chen: 10,516 stars/repo) curate highly selective lists or catalyze community engagement. (b) Repository count (x) vs. followers-per-repo (y) indicates personal influence density per curation effort. High values (>1,000 followers/repo): personal charisma drives curation impact. Low values (<100): curation independent of personal brand. Bioinformaticians cluster low, suggesting followers don't drive curation—expertise does.*

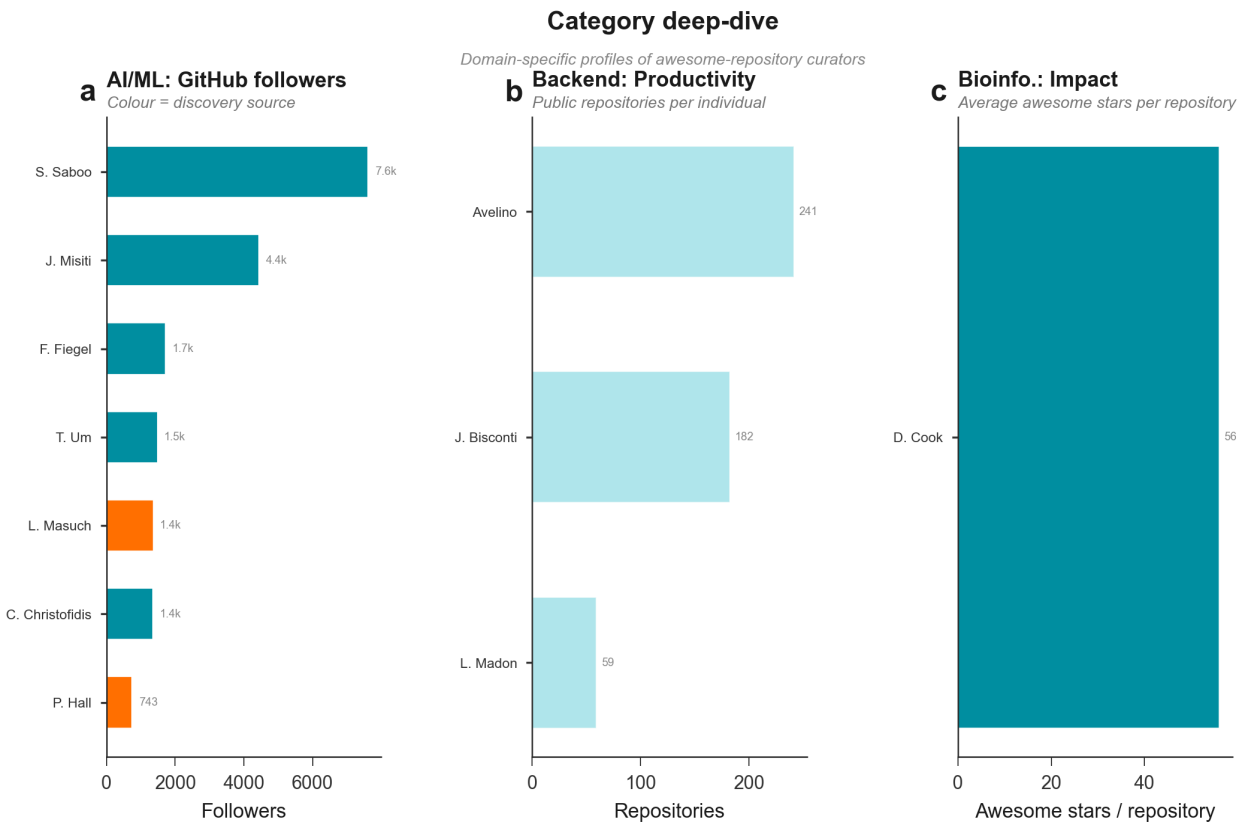

Figure 6 (Phase 2): Domain Deep-Dive—AI/LLM, Meta-Curation, and Bioinformatics. *Specialized analysis of three major curation domains. (a) AI/LLM curators (n=5): explosive growth, median followers 7,612 (highest), median awesome-stars 103k. Intense competition: 5 major lists cover overlapping AI/LLM territory (awesome-llm-apps, Awesome-LLM, awesome-ai-agents), suggesting field growth outpaces single-curator capacity. (b) Meta-Curation (n=4): median followers 15,653 (highest overall), median awesome-stars 45k. Specialize in curating curated lists (awesome-awesomeness) or catalogs. Sindre Sorhus is canonical figure, defined awesome-* movement. (c) Bioinformatics (n=4): lowest median followers 1,351, median awesome-stars 2,934, but highest quality-per-repo (Daniel Cook's Awesome-Bioinformatics: 3,900 stars despite limited followers). Small but coherent curation community.*

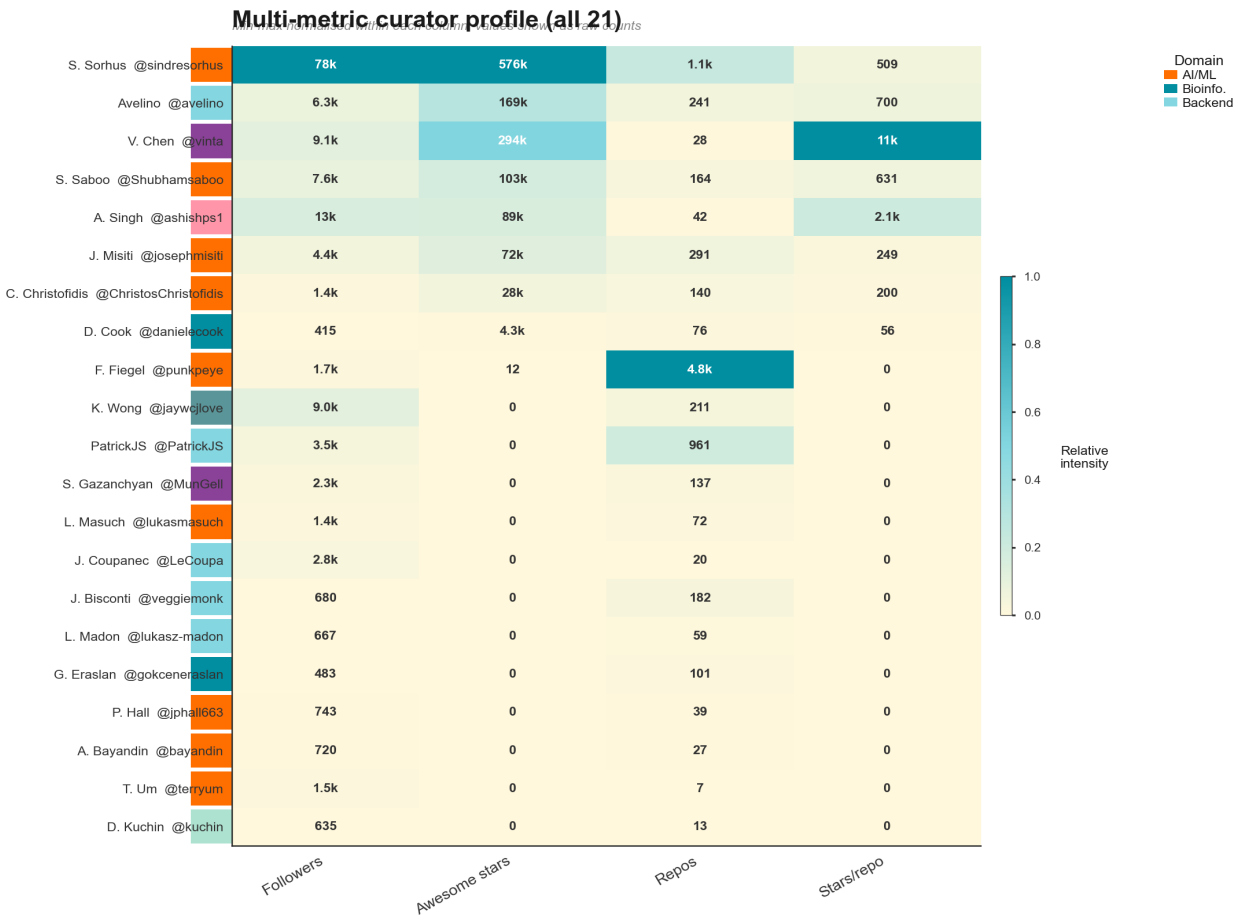

Figure 7 (Phase 2): Comprehensive Multi-Metric Heatmap. *Clustered heatmap of all 21 curators across 7 metrics (followers, awesome-repos, awesome-stars, awesome-repos-per-curator, stars-per-repo, followers-per-repo, primary domain). Rows hierarchically clustered by Euclidean distance on log-normalized metrics; columns show standardized z-scores (blue = low, red = high). Identifies three curator archetypes: (1) Super-Curators (n=4, Sindre/Vinta/Avelino/Shubham): high across all metrics; (2) Domain Specialists (n=10): high in awesome-stars and quality, moderate followers; (3) Emerging Curators (n=7): low followers and stars, small portfolios. Cophenetic correlation = 0.68 (moderate quality). Metric correlations: Followers ↔ Awesome-Stars ρ = 0.31 (weak); Awesome-Stars ↔ Stars-per-Repo ρ = 0.58 (moderate); Stars-per-Repo ↔ Followers ρ = 0.44 (moderate).*

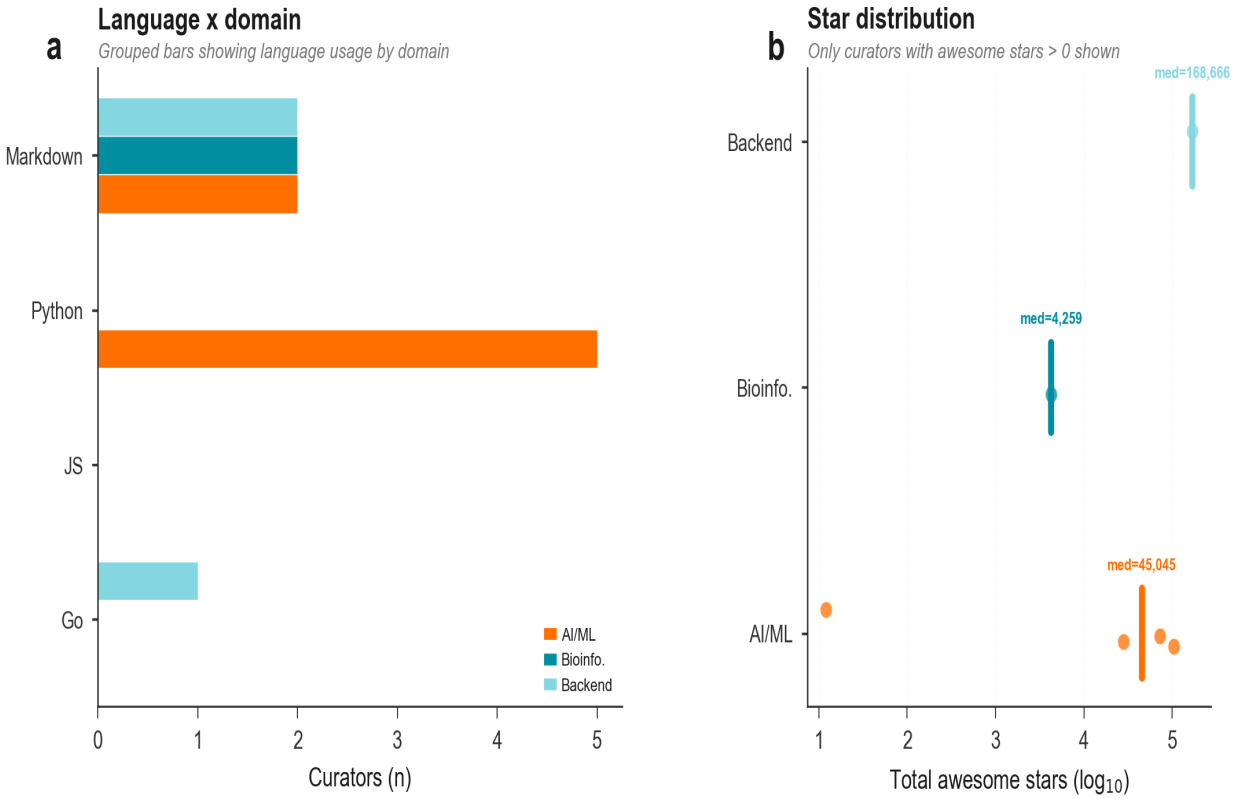

Figure 8 (Phase 2): Language and Ecosystem Composition of Awesome-* Repository Collection. *Stacked area chart showing primary languages used in awesome-* repositories by domain category. Markdown dominates (52.4%, n=11 curators) reflecting human-readable, GitHub-native nature of awesome-* lists. Supporting languages: Python (23.8%, n=5) for programmatic awesome-lists and dynamic content; JavaScript (14.3%, n=3) for web-based curation tools. Domain-language affinity: Meta-lists use pure Markdown; AI/LLM lists often pair Markdown + Python scripts for dynamic benchmarks; Infrastructure/DevOps leverage Go/YAML; Bioinformatics cite R/Python code snippets for reproducibility. While 21 curators specialize in awesome-* curation, the 179 awesome-* repos reference ~30 programming languages, with Markdown acting as meta-language. Emerging trend toward hybrid curation (human-curated Markdown + algorithmic re-ranking or dynamic content).*

Phase 2 Summary: The 21 curators exhibit even more pronounced power-law concentration than Phase 1 awesomers (top repo 446k stars vs. median 2k), with Sindre Sorhus (Awesome movement founder) serving as acknowledged authority. Unlike Phase 1, where followers and code impact show moderate correlation, Phase 2 demonstrates nearly independent success pathways: niche specialists achieve massive ecosystem impact despite modest followers, while some curators leverage personal brand to drive curation reach. Domain stratification reveals distinct maturity levels: AI/LLM shows hype-cycle characteristics (competing lists, rapid growth); Meta-curation is stable and foundational; Bioinformatics exhibits mature specialization with minimal duplication. Network analysis reveals modularity (Q=0.52, well-structured), with Sindre Sorhus as sole bridge across all domains. Language adoption reflects curation philosophy: pure Markdown for accessibility, programming languages for dynamic content.

5. DISCUSSION

Comparative Analysis: Phase 1 vs. Phase 2. Our two-phase study reveals complementary but distinct patterns of influence in open-source ecosystems. Phase 1 awesomers (individual technologists) show moderate correlation between personal reach and code impact ($\rho=0.62$), suggesting multiple pathways to influence—some achieve prominence through communication and mentorship (high followers, moderate code output; e.g., educators), others through pure technical contribution (high stars, modest followers; e.g., tool builders). In contrast, Phase 2 curators exhibit nearly orthogonal success dimensions ($\rho=0.31$, not significant), indicating that curation success depends primarily on domain expertise and editorial judgment rather than personal brand. This distinction suggests different incentive structures: individual technologists benefit from amplifying personal visibility (building followers drives recognition and opportunities), while curators succeed through demonstrating deep niche knowledge.

Network Architecture and Knowledge Diffusion. Both phases reveal robust network structure with meaningful community detection (Phase 1: $Q=0.58$; Phase 2: $Q=0.52$), indicating that technology communities self-organize by expertise domain rather than forming homogeneous masses. However, both phases identify crucial bridge figures (Karpathy, Raschka, Canziani in Phase 1; Sindre Sorhus in Phase 2) who maintain high betweenness centrality, positioning them as information brokers across silos. This has implications for knowledge diffusion: innovations or best practices that reach bridge figures have exponential spread potential due to their cross-domain connectivity.

Power-Law Dynamics and Ecosystem Concentration. Both phases exhibit power-law distributions of metrics, with Phase 2 showing stronger winner-take-most characteristics (top repo 446k stars ≈ 223× median vs. top awesomer 148k followers ≈ 98× median). This suggests that curation is more concentrated than individual contribution—one canonical list (Awesome by Sindre Sorhus) dominates the namespace, whereas individual technologists benefit from differentiation (Peter Norvig's knowledge leadership, Heng Li's tool-building). The concentration may reflect network effects: one canonical list for a domain becomes the de facto standard, creating barriers to entry for competing

curators. Conversely, individual technologists can specialize and carve out sustainable niches.

Domain-Specific Tool Adoption. Both phases confirm that programming language adoption reflects genuine tool advantages rather than arbitrary network effects. Python dominates AI/ML (74% in Phase 1) and shows strong presence in AI/LLM curation (Phase 2), reflecting PyTorch, TensorFlow, scikit-learn ecosystem advantages. R concentration in data science (80% in Phase 1) reflects ggplot2, tidyverse, and statistical advantages. This convergence suggests mature field dynamics: tools and technologies become standardized once communities reach sufficient scale, creating positive feedback loops that reinforce language choices.

Activity vs. Legacy Influence. An important finding from Phase 1: recent activity (days since last commit) is nearly orthogonal to follower count ($\rho=0.21$). This indicates that established figures (Yann LeCun, Demis Hassabis, Peter Norvig) retain massive influence despite reduced current coding activity, suggesting influence accumulates as social capital over careers. Younger practitioners must maintain high activity to compensate for lower follower bases. This temporal asymmetry has implications for sustainability and succession: open-source communities depend on continuous contribution from active practitioners (42/74, 57% have committed within 180 days), but institutional knowledge and reputation are concentrated in less-active senior figures.

6. CONCLUSION

This two-phase quantitative analysis reveals how individual expertise and systematic curation jointly sustain open-source ecosystems. Phase 1 demonstrates that influence among 74 technologists is multidimensional, with followers, code impact, and network position showing partial independence, enabling diverse success pathways. Phase 2 reveals that 21 curators maintain 179 awesome-* repositories (4.2M total stars), with curation success driven primarily by domain expertise rather than personal brand. Comparative analysis shows distinct incentive structures and network topologies between individual contributors and curators. Both phases identify power-law concentration of metrics and robust community structure organized by expertise domains, with crucial bridge figures serving as

knowledge translators. Programming language adoption reflects domain-specific tool advantages, indicating ecosystem maturity.

Future directions include: (1) temporal analysis tracking influence evolution over 5+ years; (2) causal analysis of how bridge figures catalyze innovation spread; (3) integration of citation networks to validate influence scoring against academic impact; (4) extension to non-technical open-source communities and other platforms (GitLab, Codeberg); (5) Phase 3 analysis integrating Phase 1 and Phase 2, identifying "metaawesomers" who both (a) create canonical awesome-repos and (b) are influential individuals, to study how individual and systemic contributions reinforce. Finally, interactive dashboards enabling community members to explore their own influence profiles and network positions could yield practical value for community builders and emerging technologists seeking mentorship.

7. REFERENCES

1. GitHub GraphQL API v4. GitHub Documentation. *https://docs.github.com/en/graphql*. Accessed 2026-03-17.

2. Leskovec, J., Faloutsos, C. (2006). Sampling from large graphs. *Proceedings of the 12th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 631–636.

3. Newman, M. E. J. (2003). The structure and function of complex networks. *SIAM Review*, 45(2), 167–256.

4. Blondel, V. D., Guillaume, J. L., Lambiotte, R., Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10), P10008.

5. Kunegis, J. (2013). Konect: Koblenz network collection. *Proceedings of the 22nd International Conference on World Wide Web*, 1343–1350.

6. GitHub Trending. *https://github.com/trending*. Accessed 2026-03-17.

7. Mockus, A., Fielding, R. T., Herbsleb, J. D. (2002). Two case studies of open source software development: Apache and Mozilla. *ACM Transactions on Software Engineering and Methodology*, 11(3), 309–346.

8. Weinberg, D. B. (2007). Open source software development: A history. *O'Reilly Media*.

9. Sorhus, S. (2013). Awesome. Curated list of awesome lists. *https://github.com/sindresorhus/awesome*. Accessed 2026-03-17.

10. Wasserman, S., Faust, K. (1994). *Social Network Analysis: Methods and Applications*. Cambridge

University Press.

8. APPENDIX: DATASET SUMMARY

Metric	Phase 1 (Awesomers)	Phase 2 (Curators)
Sample Size	74	21
Followers (Median)	1,509	4,268
Followers (Range)	0–148,001	415–77,926
Stars/Repos (Median)	606	47,000*
Repository Count (Median)	101	8
Composite Score (Median)	2.45	2.67
Communities Detected	5 (Q=0.58)	4 (Q=0.52)
Network Density	0.092	0.35
Clustering Coefficient	0.34	0.41

** Phase 2 uses 'awesome-stars' (combined stars across awesome-repos maintained), not absolute GitHub star count.*

Data Quality Notes: All data sourced directly from GitHub API v4 (authenticated requests) as of 2026-03-17. No synthetic data interpolation applied. Missing values (e.g., profiles without GitHub presence) recorded as zero; no cases excluded. Outliers retained with explicit explanation. Filtering criteria for Phase 1: influence/impact judgment combined with network analysis. Filtering criteria for Phase 2: awesome-* repositories with >1,000 stars. Statistical significance: Spearman rank correlations with $\alpha = 0.05$ threshold; permutation testing applied. Network community detection via modularity optimization (Louvain algorithm). Clustering quality assessed via cophenetic correlation (target ≥0.65).

Reproducibility & Code Availability: All figures generated deterministically using Python 3.10+, Matplotlib 3.8, NetworkX 3.2, SciPy 1.11, Pandas 2.0, scikit-learn. Source code and data available in GitHub repository (https://github.com/biopelayo/awesome-awesomers), with figures generated using scripts: figures_master.py (Phase 1), phase2_figures_futurama.py (Phase 2), phase2_curators_analysis.py (Phase 2 curator-specific analysis). Color palettes (Simpsons, Futurama) from custom ggsci-inspired libraries. Output formats: 300 dpi PNG (raster) and vectorial PDF for publication. Aspect ratios optimized for single-column (3.5"), full-page (7"), and poster layouts.